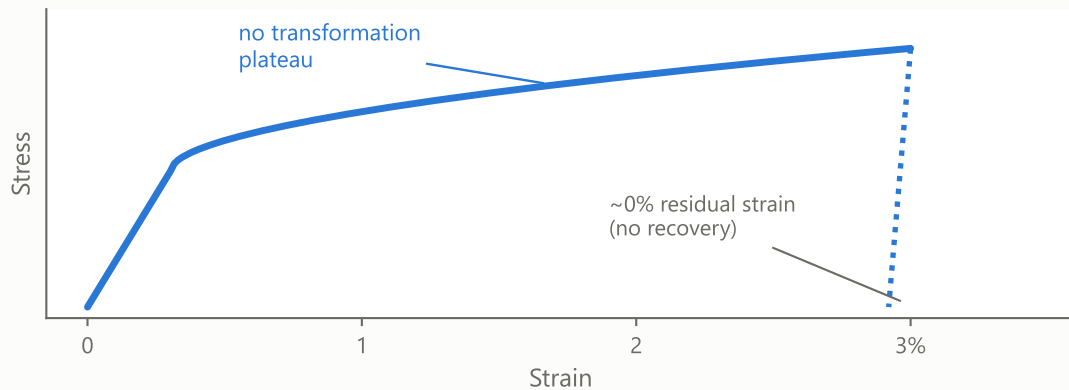


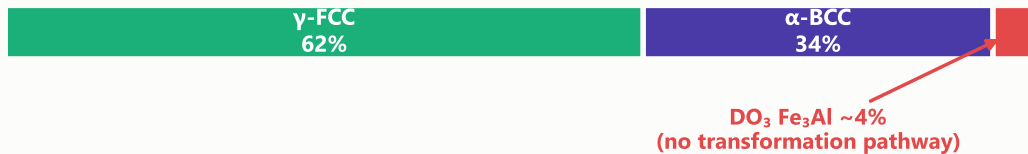
# Does an LLM-hypothesized alloy composition actually achieve super-elasticity?

## AI-HYPOTHESIZED ALLOY

*Fe-Mn-Al-Si-Ni-C (LLM deep-research agent)*



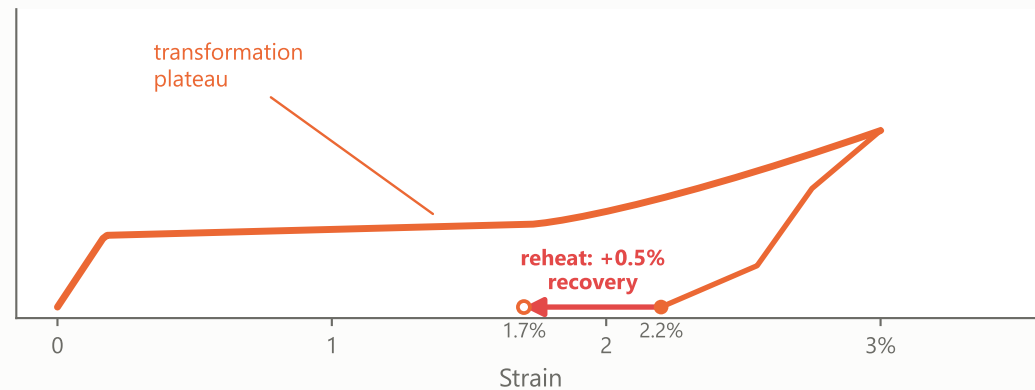
*Rietveld-refined phase fractions (synchrotron XRD)*



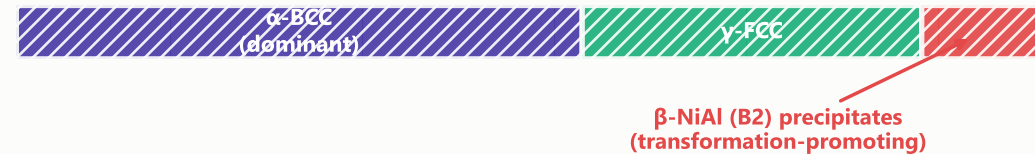
**NO super-elasticity — elastic springback only**

## BENCHMARK ALLOY

*Fe-Mn-Al-Ni (established super-elastic composition)*



*Phase constitution (qualitative; exact fractions not reported)*



**Super-elasticity via stress-induced α→γ transformation**